

1 Introduction

Real estate valuation is a complex field that is impacted by numerous variables that affect how much a property is worth. Making wise decisions in the real estate market depends critically on one's capacity to anticipate these swings. In this project, we aim to build a strong regression model that can forecast real estate values and reveal the major factors influencing property values. We focus on Sindian District, New Taipei City, Taiwan, in order to identify the major factors influencing property values.

1.1 Aim of Modelling

- 1.To build a regression model that accurately predicts house price per unit area based on the given features
- 2.To identify significant factors affecting property values.
- 3.To identify the best regression model to describe the relationship between the house price (per unit area) and its predicted regressors.

1.2 Background Information

The original owner of the real estate valuation data-set is Professor I-Cheng Yeh of Tamkang University, and it is made available by the UCI machine learning repository. It includes past real estate market data for Sindian District, New Taipei City, from 2012 to 2013. The dataset includes the following features for 414 instances: transaction date, house age, distance to the nearest MRT station, number of convenience stores, geographic coordinates (latitude and longitude), and house price per unit area (output). There are no missing values in the data.

1.3 Variable Description

Response variable (Y): House price per unit area

Predictor variables:

- X_1 : Transaction date
- X_2 : House age
- X_3 : Distance to the nearest MRT station
- X_4 : Number of convenience stores
- X_5 : Geographic coordinate (latitude)
- X_6 : Geographic coordinate (longitude)

2 Data Visualization

2.0.1 Plot of Data

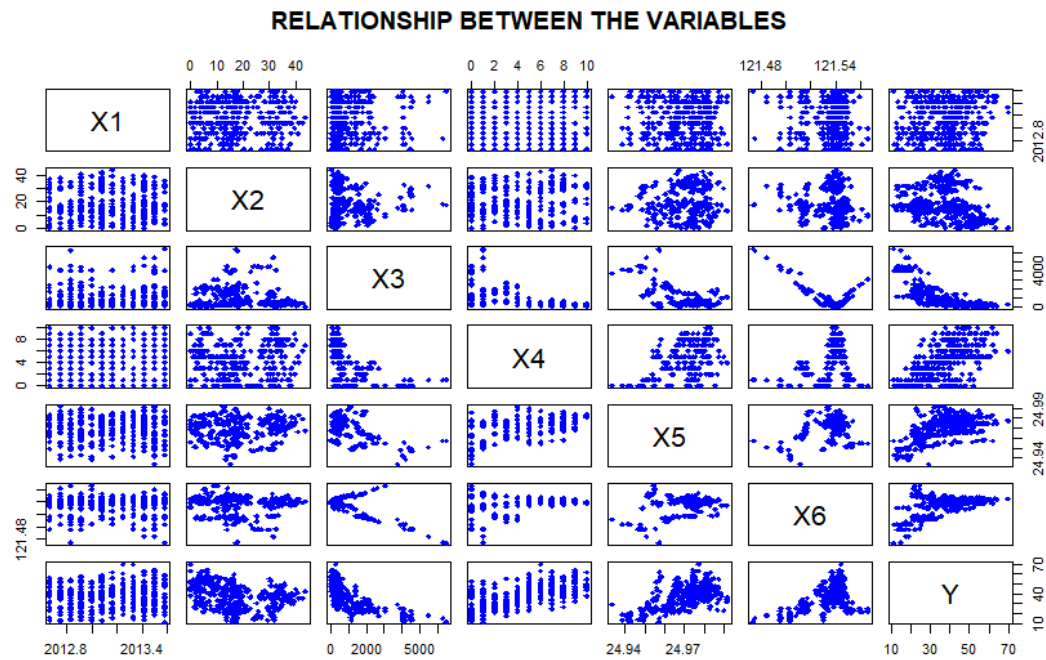


Figure 1: Plot

2.0.2 Correlation Plots

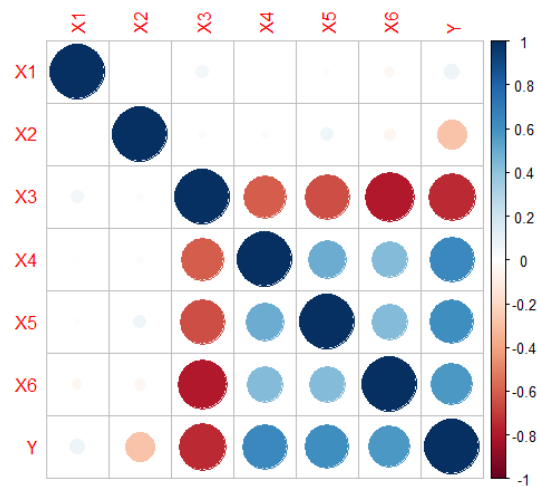


Figure 2: correlation plot 1

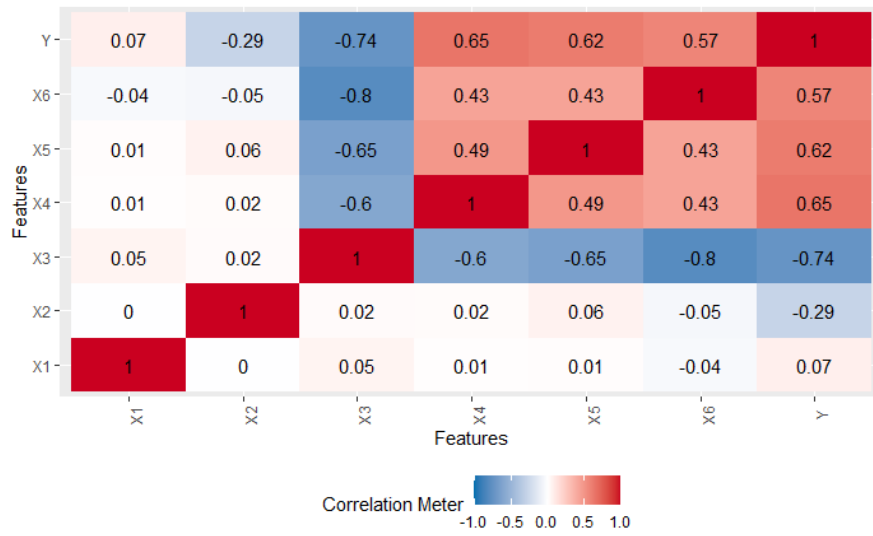


Figure 3: correlation plot 2

The correlation result above indicates that, predictor X4, X5, and X6 have a strong positive correlation with Y, predictor X1 has a weak positive correlation with Y, predictor X2 has a weak negative correlation with price and predictor X3 has a strong negative correlation with Y.

2.0.3 Histogram

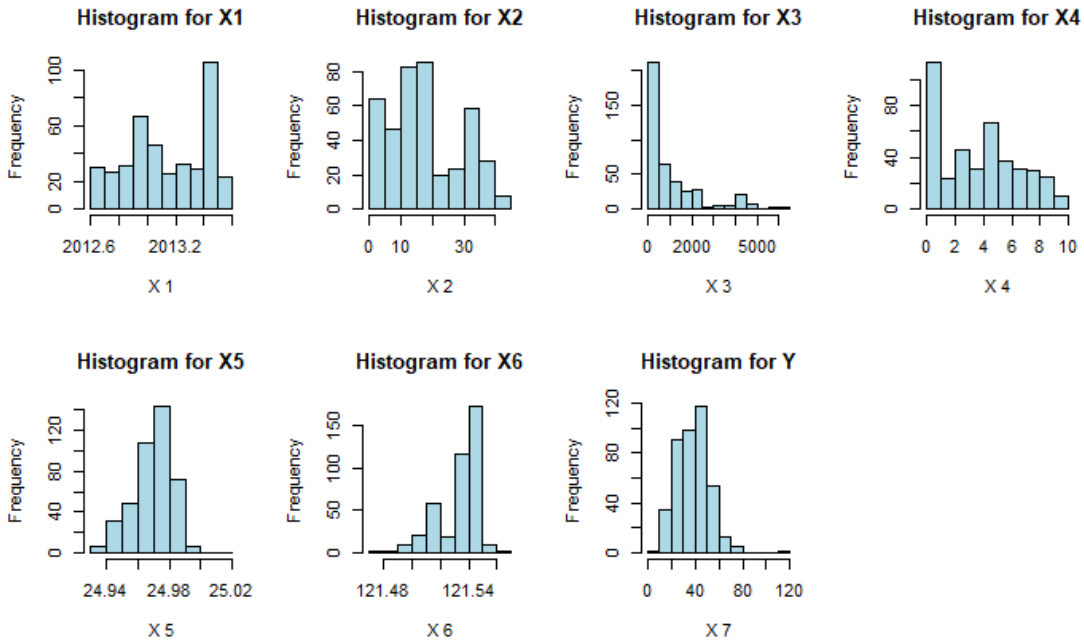


Figure 4: Histogram

2.0.4 Testing for Outliers

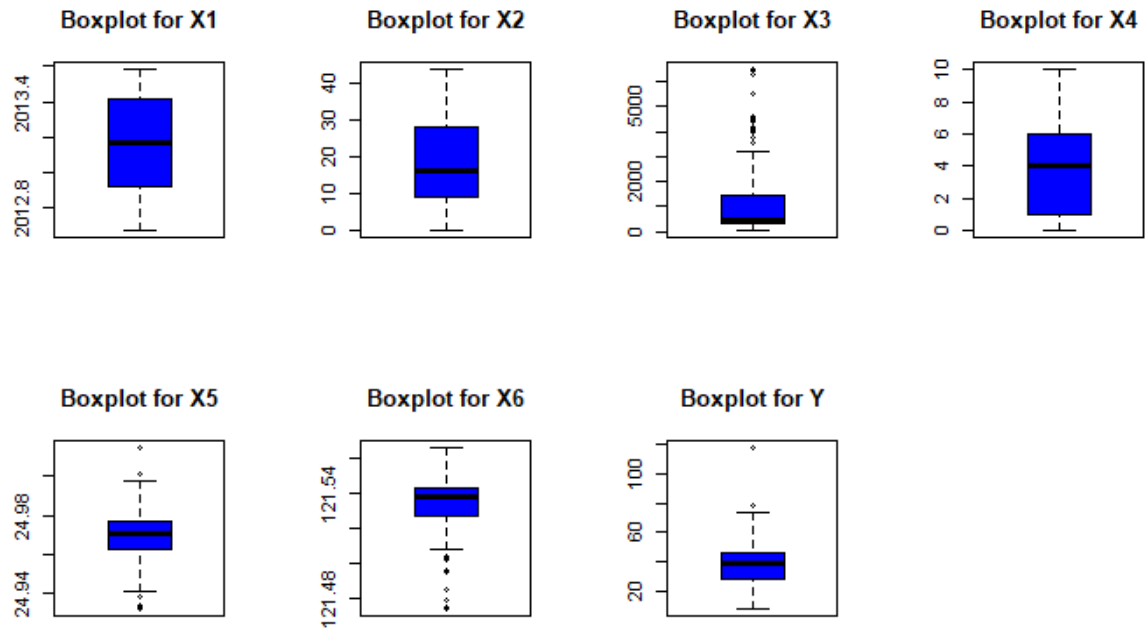


Figure 5: Boxplot

From figure 5 some of the box-plot for the variables clearly shows the presence of outliers

2.0.5 Testing for Influential Observations: Cook's Distance

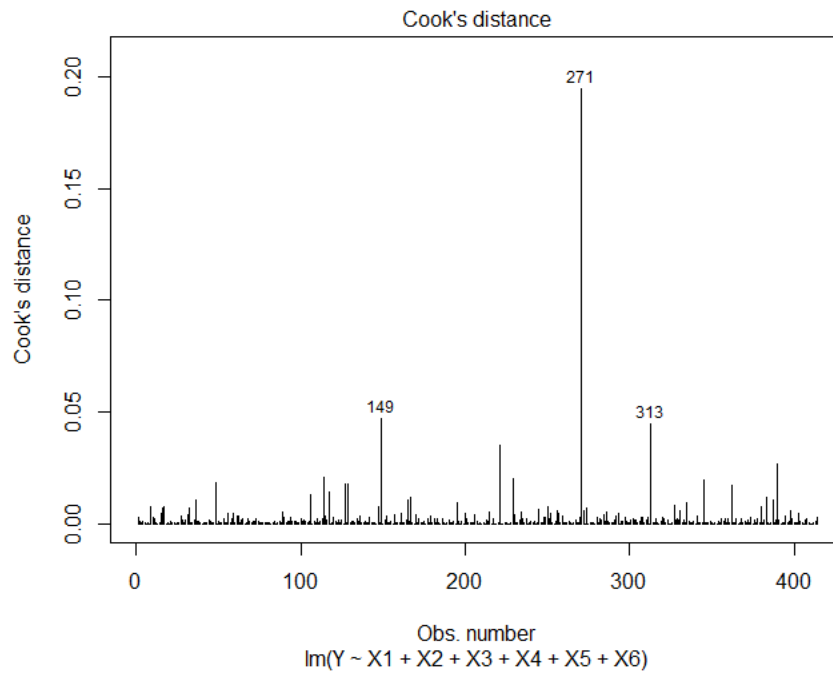


Figure 6: Cook's Distance

From figure 4, Observations 149, 271 and 313 appear to be the most influential observations in our data,

Removing influential points from the model, replacing them with a value such as the mean or median, or leaving the points in the model are some of the several approaches to handling influential points. however,in this is project I plot each variable in my data using box plots first and also I identified influential points using Cook's Distance. To find significant data points that could have a negative impact on your regression model, regression analysis uses Cook's distance, Strong influences on the fitted values are indicated by a big value for Cook's Distance for a given data point. In conclusion all these outliers are dropped and no influential point that significant impact the model was found.

2.1 Multi-Collinearity

Multi-collinearity is a statistical problem which occurs when explanatories are highly correlated with each other. the existence of collinearity inflates the variances of the estimated parameter, and mislead the relationship between the predictors and response variable, handling this I use the variance inflation factor, we identify all the VIF values over a threshold value, usually 5 Or 10.

Variables	Tolerance	VIF
X1	0.9913429	1.008733
X2	0.9879083	1.012240
X3	0.2183466	4.579874
X4	0.6151476	1.625626
X5	0.5394036	1.853899
X6	0.3472746	2.879566

Table 1: Tolerance and VIF Values

The VIF values for all predictors in our model are below the critical thresholds of 5 or 10. This indicates the absence of multicollinearity in our model.

In summary, I went through the variance inflation method to check for multi-collinearity and conclude that no problem of multicollinearity were indicated. the assumption of no collinearity holds.

2.2 Model Setup

Table 2: Model Summary

Metric	Value
R	0.974
R-Squared	0.948
Adj. R-Squared	0.944
Pred R-Squared	0.931
RMSE	3.882
Coef. Var	12.866
MSE	15.069
MAE	2.956

Table 3: ANOVA Table

Source	Sum of Squares	DF	Mean Square
Regression	7684.163	2	3842.081
Residual	421.921	28	15.069
Total	8106.084	30	

Table 4: Parameter Estimates

Model	Beta	Std. Error	Std. Beta	t	Sig.	Confidence Interval
(Intercept)	-57.988	8.638		-6.713	0.000	[-75.682, -40.293]
Height	0.339	0.130	0.132	2.607	0.014	[0.073, 0.606]
Girth	56.498	3.171	0.899	17.816	0.000	[50.002, 62.994]

Analyzing the model statistically provides insightful information. It can be seen, from the correlation coefficient ($R = 0.852$) and R-squared value (0.726) that the predictors together explain a considerable amount of the variability in the response variable. The model is statistically significant ($F = 171.134$, $p < 0.05$), as shown by the ANOVA test. X1, X4, and X5 exhibit positive connections with the response variable at the individual predictor level, but X2 and X3 show negative associations. X6 does not seem to have a statistically significant effect, which is noteworthy. The average magnitude and absolute difference between the observed and predicted values, respectively, are shown by evaluation metrics such RMSE (6.433) and MAE (5.097), which offer useful measurements of prediction accuracy. However, this does not explain the problem of overfitting, so we do the model selection by AIC to obtain the best model by selecting the model with the least AIC. The smaller the AIC, the better the fit.

2.3 Checking Assumptions

2.3.1 Fitted versus Residuals Plot

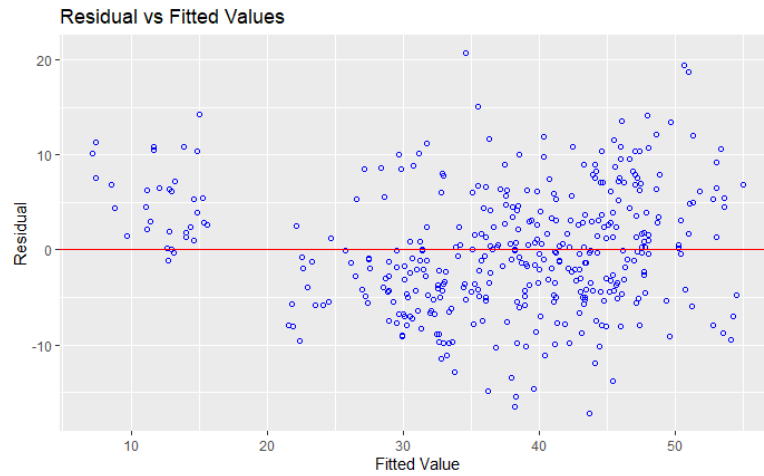


Figure 7: Residual vs Fitted Values

The residual points fall within a horizontal band centered around the 0 line, displaying no systematic pattern of positive and negative values. This shows that the assumption of linearity is valid.

2.3.2 Histogram

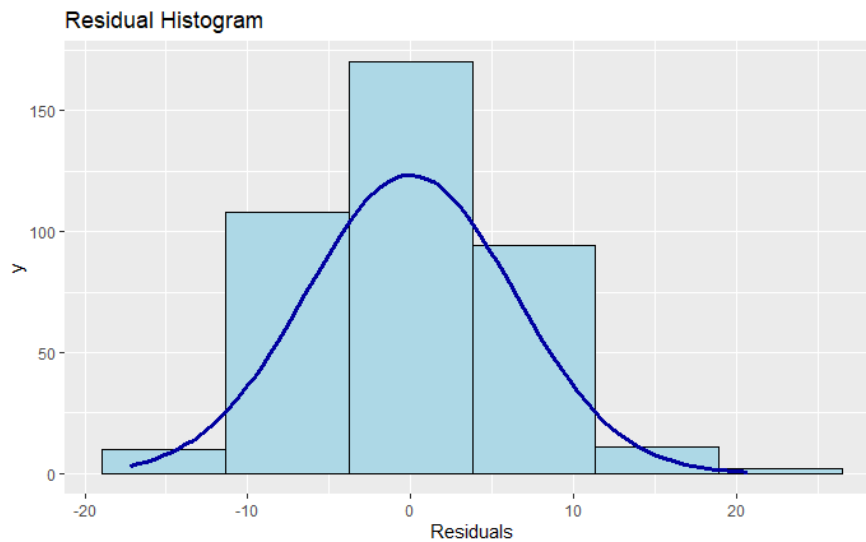


Figure 8: Histogram

The histogram appears to indicate that the distribution of the observation is approximately symmetric with a median close to zero.

2.3.3 Q-Q Plots

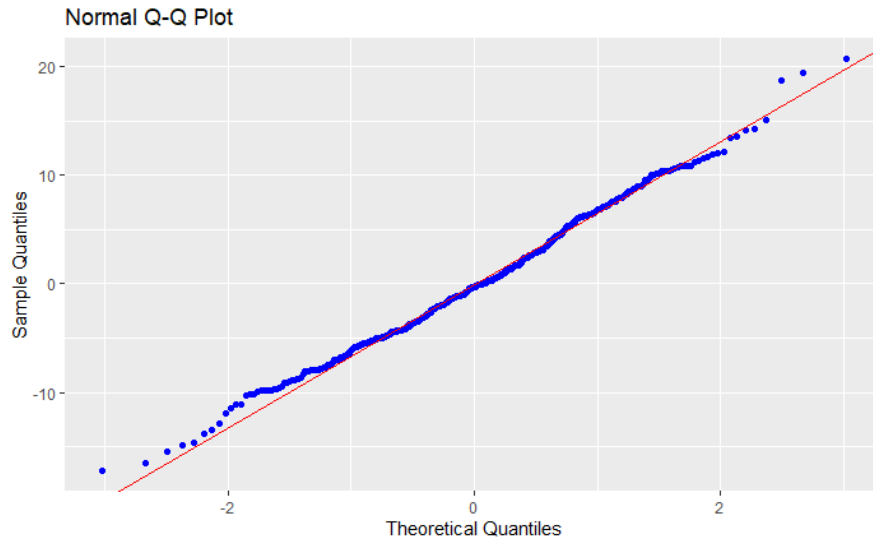


Figure 9: Normality plot

In terms of normality, the points on the normal probability Q-Q plot fall approximately on a straight line aside from the outlying point. this is an indication that the error terms are approximately normally distributed

2.3.4 Residual box plot

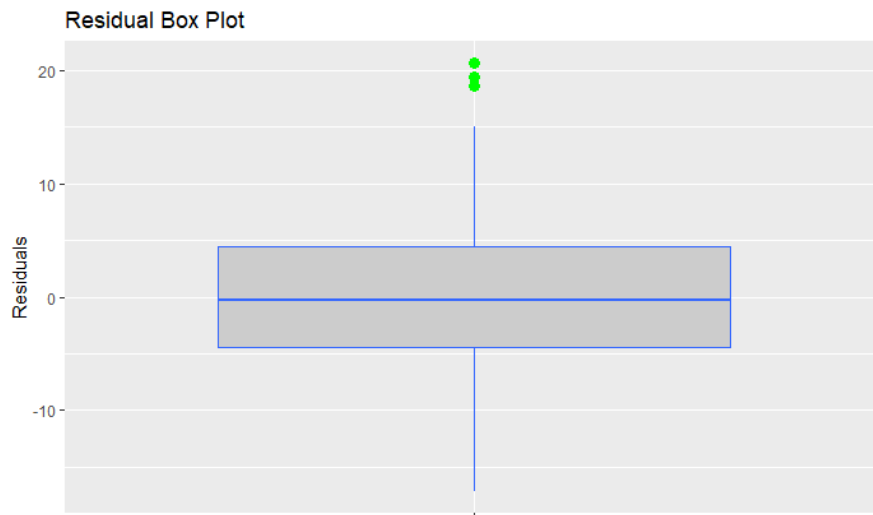


Figure 10: Boxplot

The boxplot clearly shows the presence of outliers in the data and appear to indicate that the distribution of the observation is approximately symmetric with a median close to zero.

2.3.5 Breusch-Pagan Test

Test for Heteroskedasticity	
Null Hypothesis	H_0 : The variance is constant
Alternative Hypothesis	H_a : The variance is not constant
Data	
Response	Y
Variables	Fitted values of Y
Test Summary	
DF	1
Chi-Square	1.125431
Prob > Chi-Square	0.288752

Table 5: Breusch-Pagan Test for Heteroskedasticity

Since the p-value=0.288752 which is greater than 0.05, we fail to reject the null hypothesis and conclude the assumption of constant variance is valid:

2.3.6 Shapiro-Wilk Normality Test

Shapiro-Wilk Normality Test	
Null Hypothesis	H_0 : Residuals are normally distributed
Alternative Hypothesis	H_a : Residuals are not normally distributed
Data	
Response	fit.model\$residuals
Test Summary	
W	0.99416
p-value	0.1352

Table 6: Shapiro-Wilk Normality Test for Residuals

The p-value is greater than significance level of 0.05. Hence, we fail to reject the null hypothesis and conclude that it is normally distributed.

2.4 Model Selection

Statistical models will not do well with massive irrelevant variables. therefore we ought to select a better model that only keep useful and important variables. we use the Best Subsets Regression selection techniques and select a model that has better goodness of fit, less AIC and SBIC.

Table 7: Best Subsets Regression

Model Index	Predictors
1	X_3
2	X_2, X_3
3	X_2, X_3, X_4
4	X_2, X_3, X_4, X_5
5	X_1, X_2, X_3, X_4, X_5
6	$X_1, X_2, X_3, X_4, X_5, X_6$

Table 8: Subsets Regression Summary

Model APC	R-Square	Adj. R-Square	Pred R-Square	C(p)	AIC	SBIC	SBC
1 0.4570	0.5476	0.5465	0.5428	249.0332	2788.1291	1665.3432	2800.00
2 0.3857	0.6201	0.6182	0.6134	148.4969	2721.1632	1598.4706	2737.07
3 0.3163	0.6900	0.6876	0.683	51.5902	2642.8371	1521.0608	2662.73
4 0.2893	0.7179	0.7150	0.7101	14.1373	2607.6044	1486.5390	2631.47
5 0.2828	0.7257	0.7222	0.7169	5.0825	2598.5098	1477.7616	2626.30
6 0.2841	0.7258	0.7215	0.7158	7.0000	2600.4258	1479.7163	2632.23

Different combinations of predictors (X_1 , X_2 , X_3 , X_4 , X_5 , and X_6) were thoroughly assessed using the Best Subsets Regression technique. With predictors X_1 , X_2 , X_3 , X_4 , and X_5 , the final model showed the greatest adjusted R-square (0.722). This ideal choice represents a well-balanced trade-off between model simplicity and explanatory power and was determined by criteria like the Mallows' Cp, the Schwarz Bayesian Information Criterion (SBIC), and the Akaike Information Criterion (AIC). The summary table indicates that the fifth model is the most appropriate for predicting the response variable, offering a strong basis for perceptive interpretations.

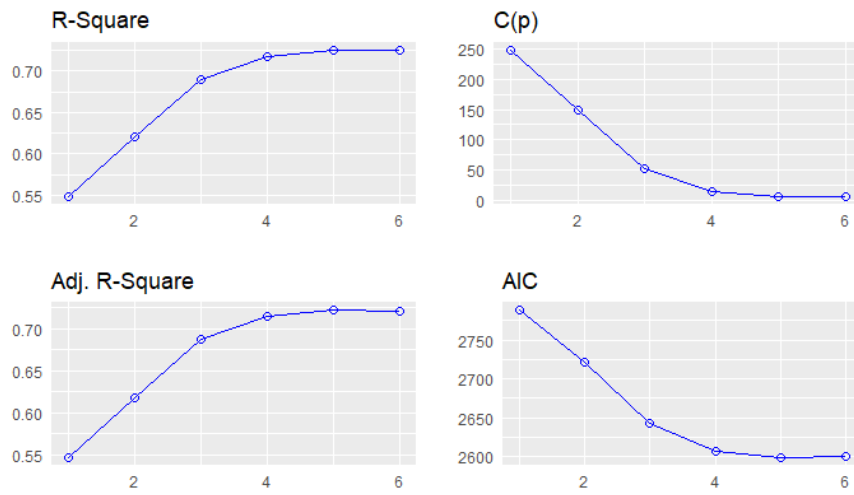


Figure 11:

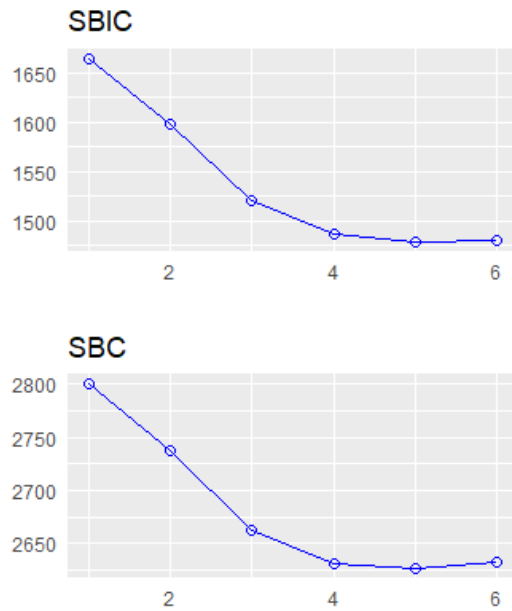


Figure 12:

2.5 Final Model Setup

Metric	Value		
R	0.852		
R-Squared	0.726		
Adj. R-Squared	0.722		
Pred R-Squared	0.717		
RMSE	6.425		
Coef. Var	17.301		
MSE	41.285		
MAE	5.090		

Metric	Value
RMSE	Root Mean Square Error
MSE	Mean Square Error
MAE	Mean Absolute Error

Table 9: Model summary metrics.

Source	Sum of Squares	DF	Mean Square	F	Sig.
Regression	42488.119	5	8497.624	205.83	0.0000
Residual	16059.751	389	41.285		
Total	58547.870	394			

Table 10: ANOVA table.

Model	Beta	Std. Error	Std. Beta	t	Sig.	Lower	Upper
(Intercept)	-13216.810	2441.041		-5.414	0.000	-18016.095	-8417.525
X1	3.829	1.150	0.089	3.329	0.001	1.567	6.090
X2	-0.324	0.029	-0.295	-11.049	0.000	-0.381	-0.266
X3	-0.004	0.000	-0.421	-10.811	0.000	-0.005	-0.003
X4	1.222	0.141	0.292	8.656	0.000	0.944	1.499
X5	222.326	36.471	0.217	6.096	0.000	150.621	294.031

Table 11: Parameter estimates.

$$Y = -13216.810 + 3.829X_1 - 0.324X_2 - 0.004X_3 + 1.222X_4 + 222.326X_5$$

The final model, incorporating predictors X1, X2, X3, X4, and X5, achieves a commendable R-Squared of 0.726, indicating its ability to explain 72.6 percent of the variability in Y. This statistically significant model ($p < 0.001$) demonstrates the individual impact of predictors: X2 and X3 negatively influence Y, while X1, X4, and X5 have positive effects. With an RMSE of 6.425 and an MAE of 5.090, the model exhibits accurate predictive performance.

2.6 Final Model Adequacy Checking

2.6.1 Fitted versus Residuals Plot

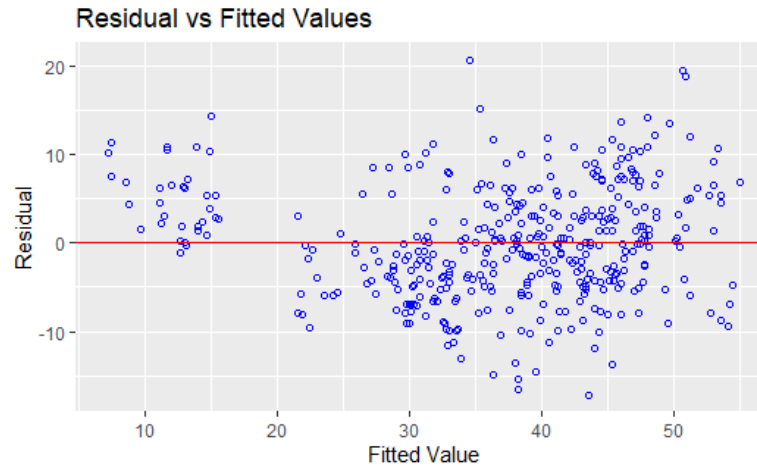


Figure 13: Residual vs Fitted Values

The residual points fall within a horizontal band centered around the 0 line, displaying no systematic pattern of positive and negative values. This shows that the assumption of linearity is valid.

2.6.2 Q-Q Plots

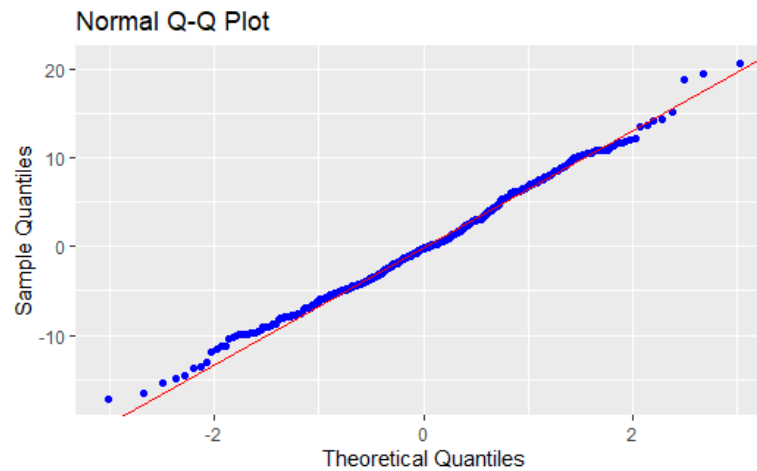


Figure 14: Normality plot

In terms of normality, the points on the normal probability Q-Q plot fall approximately on a straight line aside from the outlying point. This is an indication that the error terms are approximately normally distributed.

2.6.3 Breusch-Pagan Test

Test for Heteroskedasticity	
Null Hypothesis	H_0 : The variance is constant
Alternative Hypothesis	H_a : The variance is not constant
Data	
Response	Y
Variables	Fitted values of Y
Test Summary	
DF	1
Chi-Square	1.181926
Prob > Chi-Square	0.2769642

Table 12: Breusch-Pagan Test for Heteroskedasticity

Since the p-value= 0.2769642 which is greater than 0.05, we fail to reject the null hypothesis and conclude the assumption of constant variance is valid:

2.6.4 Shapiro-Wilk Normality Test

Shapiro-Wilk Normality Test	
Null Hypothesis	H_0 : Residuals are normally distributed
Alternative Hypothesis	H_a : Residuals are not normally distributed
Data	
Response	final.model residuals
Test Summary	
W	0.99445
p-value	0.1633

Table 13: Shapiro-Wilk Normality Test for Residuals

The p-value is greater than significance level of 0.05. Hence, we fail to reject the null hypothesis and conclude that it is normally distributed.

2.7 Application, Limitation and Disadvantages

Applications for the model include individual transactions, market trend analysis, urban planning, and real estate investment decisions. However, there are certain drawbacks, such as reliance on high-quality data, vulnerability to market fluctuations, and limited applicability. In order to maintain accuracy in navigating the ever-changing Sindian District real estate market, the project prioritizes ongoing updates and refinement in response to issues like overfitting.

2.8 Conclusion

This project aimed to develop a predictive model for real estate valuation in Sindian District, Taiwan, utilizing linear regression. Through a thorough analysis and model selection process, the final model, incorporating predictors such as transaction date, house age, distance to the nearest MRT station, number of convenience stores, and latitude, demonstrated strong explanatory power with an R-Squared of 0.726. The statistical significance of the model, assessed through ANOVA, revealed its efficacy in explaining price variability.

Additionally, the model underwent rigorous diagnostics, including checks for multicollinearity, linearity, normality, and heteroskedasticity. These analyses confirmed the model's validity and adherence to regression assumptions. The Best Subsets Regression technique facilitated the selection of the most appropriate model, striking a balance between simplicity and explanatory capacity.

In summary, the most appropriate model is

$$Y = -13216.810 + 3.829X_1 - 0.324X_2 - 0.004X_3 + 1.222X_4 + 222.326X_5$$

The house price per unit area demonstrates the following dependencies: it increases by 3.829 for each transaction date, decreases by 0.324 for every additional year in house age, decreases by 0.004 for every additional meter to the nearest MRT station, increases by 1.222 for every additional convenience store, and increases by 222.326 for every change in the latitude coordinate.